

```
In [2]:
```

```
import pandas as pd
```

```
In [3]:
```

```
df = pd.read_csv(r'Sample.csv')
```

```
df
```

	Row	Order	Order	Ship	Ship	Customer	Customer		Segment	Country/Region		Postal	
	ID	ID	Date	Date	Mode	ID	Name				City...	Code	Re
0	1	US- 2019- 103800	03- 01- 2019	07- 01- 2019	Standard Class	DP-13000	Darren Powers		Consumer	United States	Houston	...77095	Ce
1	2	US- 2019- 112326	04- 01- 2019	08- 01- 2019	Standard Class	PO-19195	Phillina Ober		Home Office	United States	Naperville	...60540	Ce
2	3	US- 2019- 112326	04- 01- 2019	08- 01- 2019	Standard Class	PO-19195	Phillina Ober		Home Office	United States	Naperville	...60540	Ce
3	4	US- 2019- 112326	04- 01- 2019	08- 01- 2019	Standard Class	PO-19195	Phillina Ober		Home Office	United States	Naperville	...60540	Ce
4	5	US- 2019- 141817	05- 01- 2019	12- 01- 2019	Standard Class	MB- 18085	Mick Brown		Consumer	United States	Philadelphia	...19143	Ea
...	...	...	...	...	...	...	...	...	...	...	...	...	...
10189	10190	US- 2022- 143259	30- 12- 2022	03- 01- 2023	Standard Class	PO-18865	Patrick O'Donnell		Consumer	United States	New York City	...10009	Ea
10190	10191	US- 2022- 115427	30- 12- 2022	03- 01- 2023	Standard Class	EB-13975	Erica Bern		Corporate	United States	Fairfield	...94533	We
10191	10192	US- 2022- 156720	30- 12- 2022	03- 01- 2023	Standard Class	JM-15580	Jill Matthias		Consumer	United States	Loveland	...80538	We
10192	10193	US- 2022- 143259	30- 12- 2022	03- 01- 2023	Standard Class	PO-18865	Patrick O'Donnell		Consumer	United States	New York City	...10009	Ea
10193	10194	CA- 2022- 143500	30- 12- 2022	03- 01- 2023	Standard Class	HO- 15230	Harry Olson		Consumer	Canada	Charlottetown...	...C0A	Ea
10194 rows × 21 columns													

	Row ID	Order ID	Order Date	Ship Date	Ship Mode \
0	1	US-2019-103800	03-01-2019	07-01-2019	Standard Class
1	2	US-2019-112326	04-01-2019	08-01-2019	Standard Class
2	3	US-2019-112326	04-01-2019	08-01-2019	Standard Class
3	4	US-2019-112326	04-01-2019	08-01-2019	Standard Class
4	5	US-2019-141817	05-01-2019	12-01-2019	Standard Class
...	...	...	...	...	...
10189	10190	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10190	10191	US-2022-115427	30-12-2022	03-01-2023	Standard Class
10191	10192	US-2022-156720	30-12-2022	03-01-2023	Standard Class
10192	10193	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10193	10194	CA-2022-143500	30-12-2022	03-01-2023	Standard Class

	Customer ID	Customer Name	Segment	Country/Region \
0	DP-13000	Darren Powers	Consumer	United States
1	PO-19195	Phillina Ober	Home Office	United States
2	PO-19195	Phillina Ober	Home Office	United States
3	PO-19195	Phillina Ober	Home Office	United States
4	MB-18085	Mick Brown	Consumer	United States
...	...	...	...	...
10189	PO-18865	Patrick O'Donnell	Consumer	United States
10190	EB-13975	Erica Bern	Corporate	United States
10191	JM-15580	Jill Matthias	Consumer	United States
10192	PO-18865	Patrick O'Donnell	Consumer	United States
10193	HO-15230	Harry Olson	Consumer	Canada

	City	Postal Code	Region	Product ID \
0	Houston	77095	Central	OFF-PA-10000174
1	Naperville	60540	Central	OFF-BI-10004094
2	Naperville	60540	Central	OFF-LA-10003223
3	Naperville	60540	Central	OFF-ST-10002743
4	Philadelphia	19143	East	OFF-AR-10003478
...	...	...	...	...
10189	New York City	10009	East	OFF-BI-10003684
10190	Fairfield	94533	West	OFF-BI-10004632
10191	Loveland	80538	West	OFF-FA-10003472
10192	New York City	10009	East	TEC-PH-10004774
10193	Charlottetown	C0A	East	OFF-BI-10004040

	Category	Sub-Category \
0	Office Supplies	Paper
1	Office Supplies	Binders
2	Office Supplies	Labels
3	Office Supplies	Storage
4	Office Supplies	Art
...	...	...
10189	Office Supplies	Binders
10190	Office Supplies	Binders
10191	Office Supplies	Fasteners
10192	Technology	Phones
10193	Office Supplies	Binders

	Product Name	Sales	Quantity \
0	Message Book, Wirebound, Four 5 1/2" X 4" Form...	16.448	2
1	GBC Standard Plastic Binding Systems Combs	3.540	2
2	Avery 508	11.784	3
3	SAFCO Boltless Steel Shelving	272.736	3
4	Avery Hi-Liter EverBold Pen Style Fluorescent ...	19.536	3
...	...	...	...
10189	Wilson Jones Legal Size Ring Binders	52.776	3
10190	GBC Binding covers	20.720	2
10191	Bagged Rubber Bands	3.024	3
10192	Gear Head AU3700S Headset	90.930	7
10193	Wilson Jones Impact Binders	3.024	3

	Discount	Profit
0	0.2	5.5512
1	0.8	-5.4870
2	0.2	4.2717

```
3          0.2 -64.7748
4          0.2  4.8840
...        ...      ...
10189      0.2 19.7910
10190      0.2  6.4750
10191      0.2 -0.6048
10192      0.0  2.7279
10193      0.2 -0.6048
```

```
[10194 rows x 21 columns]
```

```
In [4]:
```

```
# 1. Write a function that returns all orders where Sales > 500.
```

```
def high_sales_orders(df):
    return df[df['Sales'] > 500]
```

```
print(high_sales_orders(df))
```

stdout:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode \
5	6	US-2019-167199	06-01-2019	10-01-2019	Standard Class
8	9	US-2019-167199	06-01-2019	10-01-2019	Standard Class
12	13	US-2019-167199	06-01-2019	10-01-2019	Standard Class
22	23	US-2019-109232	13-01-2019	16-01-2019	Second Class
29	30	US-2019-162775	13-01-2019	15-01-2019	Second Class
...	...	...	...	...	...
10121	10122	US-2022-145219	24-12-2022	25-12-2022	First Class
10137	10138	US-2022-114055	25-12-2022	29-12-2022	Second Class
10149	10150	US-2022-121741	26-12-2022	26-12-2022	Same Day
10169	10170	US-2022-163979	28-12-2022	02-01-2023	Second Class
10172	10173	US-2022-158526	29-12-2022	01-01-2023	Second Class

	Customer ID	Customer Name	Segment	Country/Region \
5	ME-17320	Maria Etezadi	Home Office	United States
8	ME-17320	Maria Etezadi	Home Office	United States
12	ME-17320	Maria Etezadi	Home Office	United States
22	ND-18370	Natalie DeCherney	Consumer	United States
29	CS-12250	Chris Selesnick	Corporate	United States
...	...	...	...	...
10121	RM-19675	Robert Marley	Home Office	United States
10137	MH-18115	Mick Hernandez	Home Office	United States
10149	YC-21895	Yoseph Carroll	Corporate	United States
10169	KH-16690	Kristen Hastings	Corporate	United States
10172	KH-16360	Katherine Hughes	Consumer	United States

	City	Postal Code	Region	Product ID \
5	Henderson	42420	South	FUR-CH-10004063
8	Henderson	42420	South	OFF-BI-10004632
12	Henderson	42420	South	TEC-PH-10004539
22	Mount Pleasant	29464	South	FUR-CH-10000422
29	Bossier City	71111	South	OFF-ST-10000025
...	...	...	...	...
10121	Los Angeles	90049	West	TEC-CO-10001449
10137	Huntsville	35810	South	OFF-PA-10000994
10149	Fremont	68025	Central	OFF-ST-10004459
10169	San Francisco	94110	West	OFF-ST-10003208
10172	Louisville	40214	South	FUR-CH-10002602

	Category	Sub-Category \
5	Furniture	Chairs
8	Office Supplies	Binders
12	Technology	Phones
22	Furniture	Chairs
29	Office Supplies	Storage
...	...	...
10121	Technology	Copiers
10137	Office Supplies	Paper
10149	Office Supplies	Storage
10169	Office Supplies	Storage
10172	Furniture	Chairs

	Product Name	Sales	Quantity \
5	Global Deluxe High-Back Manager's Chair	2573.820	9
8	Ibico Hi-Tech Manual Binding System	609.980	2
12	Wireless Extenders zBoost YX545 SOHO Signal Bo...	755.960	4
22	Global Highback Leather Tilter in Burgundy	545.940	6
29	Fellowes Stor/Drawer Steel Plus Storage Drawers	572.580	6
...	...	...	...
10121	Hewlett Packard LaserJet 3310 Copier	2879.952	6
10137	Xerox 1915	629.100	6
10149	Tennsco Single-Tier Lockers	750.680	2
10169	Adjustable Depth Letter/Legal Cart	725.840	4
10172	DMI Arturo Collection Mission-style Design Woo...	1207.840	8

	Discount	Profit
5	0.0	746.4078
8	0.0	274.4910
12	0.0	204.1092
22	0.0	87.3504

```
29          0.0    34.3548
...          ...         ...
10121        0.2  1007.9832
10137        0.0   301.9680
10149        0.0    37.5340
10169        0.0   210.4936
10172        0.0   314.0384
```

```
[1175 rows x 21 columns]
```

```
In [7]:
```

```
# 2. Create a function to return orders where Category = 'Technology' and Profit > 100.
```

```
def tech_profit_orders(df):
    return df[(df['Category'] == 'Technology') & (df['Profit'] > 100)]
```

```
result = tech_profit_orders(df)
```

```
print(tech_profit_orders(df))
```

stdout:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode \
12	13	US-2019-167199	06-01-2019	10-01-2019	Standard Class
13	14	US-2019-167199	06-01-2019	10-01-2019	Standard Class
31	32	US-2019-162775	13-01-2019	15-01-2019	Second Class
59	60	US-2019-147627	20-01-2019	26-01-2019	Standard Class
84	85	US-2019-140795	01-02-2019	03-02-2019	First Class
...	...	...	...	...	...
9983	9984	US-2022-151799	14-12-2022	18-12-2022	Standard Class
10086	10087	US-2022-149699	22-12-2022	24-12-2022	First Class
10106	10107	US-2022-162670	23-12-2022	28-12-2022	Second Class
10121	10122	US-2022-145219	24-12-2022	25-12-2022	First Class
10122	10123	US-2022-124296	24-12-2022	28-12-2022	Standard Class

	Customer ID	Customer Name	Segment	Country/Region \
12	ME-17320	Maria Etezadi	Home Office	United States
13	ME-17320	Maria Etezadi	Home Office	United States
31	CS-12250	Chris Selesnick	Corporate	United States
59	HL-15040	Hunter Lopez	Consumer	United States
84	BD-11500	Bradley Drucker	Consumer	United States
...	...	...	...	...
9983	BF-11170	Ben Ferrer	Home Office	United States
10086	CM-12115	Chad McGuire	Consumer	United States
10106	MF-18250	Monica Federle	Corporate	United States
10121	RM-19675	Robert Marley	Home Office	United States
10122	CS-12355	Christine Sundaresam	Consumer	United States

	City	Postal Code	Region	Product ID	Category \	
12	Henderson	...	42420	South	TEC-PH-10004539	Technology
13	Henderson	...	42420	South	TEC-PH-10004977	Technology
31	Bossier City	...	71111	South	TEC-AC-10003174	Technology
59	Jonesboro	...	72401	South	TEC-PH-10001061	Technology
84	Green Bay	...	54302	Central	TEC-AC-10001432	Technology
...	...	...	...	...	...	...
9983	Lawrence	...	01841	East	TEC-CO-10002313	Technology
10086	Los Angeles	...	90008	West	TEC-AC-10000474	Technology
10106	Little Rock	...	72209	South	TEC-PH-10003601	Technology
10121	Los Angeles	...	90049	West	TEC-CO-10001449	Technology
10122	Lafayette	...	70506	South	TEC-MA-10003183	Technology

	Sub-Category	Product Name \
12	Phones	Wireless Extenders zBoost YX545 SOHO Signal Bo...
13	Phones	GE 30524EE4
31	Accessories	Plantronics S12 Corded Telephone Headset System
59	Phones	Apple iPhone 5C
84	Accessories	Enermax Aurora Lite Keyboard
...	...	...
9983	Copiers	Canon PC1080F Personal Copier
10086	Accessories	Kensington Expert Mouse Optical USB Trackball ...
10106	Phones	Ativa D5772 2-Line 5.8GHz Digital Expandable C...
10121	Copiers	Hewlett Packard LaserJet 3310 Copier
10122	Machines	DYMO CardScan Personal V9 Business Card Scanner

	Sales	Quantity	Discount	Profit
12	755.960	4	0.0	204.1092
13	391.980	2	0.0	113.6742
31	646.740	6	0.0	258.6960
59	699.930	7	0.0	181.9818
84	468.900	6	0.0	206.3160
...	...	...	...	...
9983	1199.980	2	0.0	467.9922
10086	474.950	5	0.0	142.4850
10106	494.970	3	0.0	148.4910
10121	2879.952	6	0.2	1007.9832
10122	479.970	3	0.0	239.9850

[357 rows x 21 columns]

In [8]:

```
# 3. Write a function to create a new column 'Profit_Status' where Profit > 0 → 'Profit' else 'Loss'.
```

```
def add_profit_status(df):  
    df['Profit_Status'] = 'Loss'  
    df.loc[df['Profit'] > 0, 'Profit_Status'] = 'Profit'  
    return df
```

```
print(add_profit_status(df))
```

stdout:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode \
0	1	US-2019-103800	03-01-2019	07-01-2019	Standard Class
1	2	US-2019-112326	04-01-2019	08-01-2019	Standard Class
2	3	US-2019-112326	04-01-2019	08-01-2019	Standard Class
3	4	US-2019-112326	04-01-2019	08-01-2019	Standard Class
4	5	US-2019-141817	05-01-2019	12-01-2019	Standard Class
...	...	...	...	...	...
10189	10190	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10190	10191	US-2022-115427	30-12-2022	03-01-2023	Standard Class
10191	10192	US-2022-156720	30-12-2022	03-01-2023	Standard Class
10192	10193	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10193	10194	CA-2022-143500	30-12-2022	03-01-2023	Standard Class

	Customer ID	Customer Name	Segment	Country/Region \
0	DP-13000	Darren Powers	Consumer	United States
1	PO-19195	Phillina Ober	Home Office	United States
2	PO-19195	Phillina Ober	Home Office	United States
3	PO-19195	Phillina Ober	Home Office	United States
4	MB-18085	Mick Brown	Consumer	United States
...	...	...	...	...
10189	PO-18865	Patrick O'Donnell	Consumer	United States
10190	EB-13975	Erica Bern	Corporate	United States
10191	JM-15580	Jill Matthias	Consumer	United States
10192	PO-18865	Patrick O'Donnell	Consumer	United States
10193	HO-15230	Harry Olson	Consumer	Canada

	City	...	Region	Product ID	Category \
0	Houston	...	Central	OFF-PA-10000174	Office Supplies
1	Naperville	...	Central	OFF-BI-10004094	Office Supplies
2	Naperville	...	Central	OFF-LA-10003223	Office Supplies
3	Naperville	...	Central	OFF-ST-10002743	Office Supplies
4	Philadelphia	...	East	OFF-AR-10003478	Office Supplies
...	...	...	...	...	...
10189	New York City	...	East	OFF-BI-10003684	Office Supplies
10190	Fairfield	...	West	OFF-BI-10004632	Office Supplies
10191	Loveland	...	West	OFF-FA-10003472	Office Supplies
10192	New York City	...	East	TEC-PH-10004774	Technology
10193	Charlottetown	...	East	OFF-BI-10004040	Office Supplies

	Sub-Category	Product Name \
0	Paper	Message Book, Wirebound, Four 5 1/2" X 4" Form...
1	Binders	GBC Standard Plastic Binding Systems Combs
2	Labels	Avery 508
3	Storage	SAFCO Boltless Steel Shelving
4	Art	Avery Hi-Liter EverBold Pen Style Fluorescent ...
...	...	...
10189	Binders	Wilson Jones Legal Size Ring Binders
10190	Binders	GBC Binding covers
10191	Fasteners	Bagged Rubber Bands
10192	Phones	Gear Head AU3700S Headset
10193	Binders	Wilson Jones Impact Binders

	Sales Quantity	Discount	Profit	Profit_Status
0	16.448	2	0.2	5.5512 Profit
1	3.540	2	0.8	-5.4870 Loss
2	11.784	3	0.2	4.2717 Profit
3	272.736	3	0.2	-64.7748 Loss
4	19.536	3	0.2	4.8840 Profit
...	...	...	...	...
10189	52.776	3	0.2	19.7910 Profit
10190	20.720	2	0.2	6.4750 Profit
10191	3.024	3	0.2	-0.6048 Loss
10192	90.930	7	0.0	2.7279 Profit
10193	3.024	3	0.2	-0.6048 Loss

[10194 rows x 22 columns]

```
In [9]:

# 4. Write a function to return total sales per region.

def total_sales_region(df):
    return df.groupby('Region')['Sales'].sum()

print(total_sales_region(df))
```

```
stdout:
Region
Central    503170.6728
East       691828.1680
South      391721.9050
West       739813.6085
Name: Sales, dtype: float64
```

```
In [12]:

# 5. Write a function to return top 5 highest profit orders.

def top5_profit_orders(df):
    return df.sort_values(by='Profit', ascending=False).head(5)

print(top5_profit_orders(df))
```

```
stdout:

   Row ID   Order ID  Order Date  Ship Date  Ship Mode \
5894   5895  US-2021-118689  02-10-2021  09-10-2021  Standard Class
7255   7256  US-2022-140151  23-03-2022  25-03-2022    First Class
9517   9518  US-2022-166709  17-11-2022  22-11-2022  Standard Class
6674   6675  US-2021-117121  17-12-2021  21-12-2021  Standard Class
1203   1204  US-2019-116904  23-09-2019  28-09-2019  Standard Class

   Customer ID  Customer Name  Segment  Country/Region  City  ... \
5894   TC-20980   Tamara Chand  Corporate  United States  Lafayette  ...
7255   RB-19360   Raymond Buch  Consumer  United States  Seattle  ...
9517   HL-15040   Hunter Lopez  Consumer  United States  Newark  ...
6674   AB-10105   Adrian Barton  Consumer  United States  Detroit  ...
1203   SC-20095   Sanjit Chand  Consumer  United States  Minneapolis  ...

   Region  Product ID  Category  Sub-Category \
5894  Central  TEC-CO-10004722  Technology  Copiers
7255   West  TEC-CO-10004722  Technology  Copiers
9517   East  TEC-CO-10004722  Technology  Copiers
6674  Central  OFF-BI-10000545  Office Supplies  Binders
1203  Central  OFF-BI-10001120  Office Supplies  Binders

   Product Name  Sales Quantity \
5894  Canon imageCLASS 2200 Advanced Copier  17499.95  5
7255  Canon imageCLASS 2200 Advanced Copier  13999.96  4
9517  Canon imageCLASS 2200 Advanced Copier  10499.97  3
6674  GBC Ibimaster 500 Manual ProClick Binding System  9892.74  13
1203  Ibico EPK-21 Electric Binding System  9449.95  5

   Discount  Profit  Profit_Status
5894      0.0  8399.9760      Profit
7255      0.0  6719.9808      Profit
9517      0.0  5039.9856      Profit
6674      0.0  4946.3700      Profit
1203      0.0  4630.4755      Profit
```

```
[5 rows x 22 columns]
```

In [13]:

6. Write a function to fill missing values in 'Postal Code' with 0.

```
def fill_postal_code(df):  
    df['Postal Code'] = df['Postal Code'].fillna(0)  
    return df  
  
print(fill_postal_code(df))
```

stdout:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode \
0	1	US-2019-103800	03-01-2019	07-01-2019	Standard Class
1	2	US-2019-112326	04-01-2019	08-01-2019	Standard Class
2	3	US-2019-112326	04-01-2019	08-01-2019	Standard Class
3	4	US-2019-112326	04-01-2019	08-01-2019	Standard Class
4	5	US-2019-141817	05-01-2019	12-01-2019	Standard Class
...	...	...	...	...	...
10189	10190	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10190	10191	US-2022-115427	30-12-2022	03-01-2023	Standard Class
10191	10192	US-2022-156720	30-12-2022	03-01-2023	Standard Class
10192	10193	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10193	10194	CA-2022-143500	30-12-2022	03-01-2023	Standard Class

	Customer ID	Customer Name	Segment	Country/Region \
0	DP-13000	Darren Powers	Consumer	United States
1	PO-19195	Phillina Ober	Home Office	United States
2	PO-19195	Phillina Ober	Home Office	United States
3	PO-19195	Phillina Ober	Home Office	United States
4	MB-18085	Mick Brown	Consumer	United States
...	...	...	...	...
10189	PO-18865	Patrick O'Donnell	Consumer	United States
10190	EB-13975	Erica Bern	Corporate	United States
10191	JM-15580	Jill Matthias	Consumer	United States
10192	PO-18865	Patrick O'Donnell	Consumer	United States
10193	HO-15230	Harry Olson	Consumer	Canada

	City	...	Region	Product ID	Category \
0	Houston	...	Central	OFF-PA-10000174	Office Supplies
1	Naperville	...	Central	OFF-BI-10004094	Office Supplies
2	Naperville	...	Central	OFF-LA-10003223	Office Supplies
3	Naperville	...	Central	OFF-ST-10002743	Office Supplies
4	Philadelphia	...	East	OFF-AR-10003478	Office Supplies
...	...	...	...	...	...
10189	New York City	...	East	OFF-BI-10003684	Office Supplies
10190	Fairfield	...	West	OFF-BI-10004632	Office Supplies
10191	Loveland	...	West	OFF-FA-10003472	Office Supplies
10192	New York City	...	East	TEC-PH-10004774	Technology
10193	Charlottetown	...	East	OFF-BI-10004040	Office Supplies

	Sub-Category	Product Name \
0	Paper	Message Book, Wirebound, Four 5 1/2" X 4" Form...
1	Binders	GBC Standard Plastic Binding Systems Combs
2	Labels	Avery 508
3	Storage	SAFCO Boltless Steel Shelving
4	Art	Avery Hi-Liter EverBold Pen Style Fluorescent ...
...	...	...
10189	Binders	Wilson Jones Legal Size Ring Binders
10190	Binders	GBC Binding covers
10191	Fasteners	Bagged Rubber Bands
10192	Phones	Gear Head AU3700S Headset
10193	Binders	Wilson Jones Impact Binders

	Sales Quantity	Discount	Profit	Profit_Status
0	16.448	2	0.2	5.5512 Profit
1	3.540	2	0.8	-5.4870 Loss
2	11.784	3	0.2	4.2717 Profit
3	272.736	3	0.2	-64.7748 Loss
4	19.536	3	0.2	4.8840 Profit
...	...	...	...	...
10189	52.776	3	0.2	19.7910 Profit
10190	20.720	2	0.2	6.4750 Profit
10191	3.024	3	0.2	-0.6048 Loss
10192	90.930	7	0.0	2.7279 Profit
10193	3.024	3	0.2	-0.6048 Loss

[10194 rows x 22 columns]

```

In [18]:

# 7. Write a function to apply discount rule: If Sales > 1000 → 20% discount else → 10%.

def apply_discount(df):
    df['Discount'] = 0.0
    df.loc[df['Sales'] > 1000, 'Discount'] = df['Sales'] * 0.20
    df.loc[df['Sales'] <= 1000, 'Discount'] = df['Sales'] * 0.10
    return df

df = apply_discount(df)

print(df.head())

```

```

stdout:

   Row ID  Order ID  Order Date  Ship Date  Ship Mode  Customer ID \
0      1  US-2019-103800  03-01-2019  07-01-2019  Standard Class  DP-13000
1      2  US-2019-112326  04-01-2019  08-01-2019  Standard Class  PO-19195
2      3  US-2019-112326  04-01-2019  08-01-2019  Standard Class  PO-19195
3      4  US-2019-112326  04-01-2019  08-01-2019  Standard Class  PO-19195
4      5  US-2019-141817  05-01-2019  12-01-2019  Standard Class  MB-18085

   Customer Name  Segment  Country/Region  City  ...  Region \
0  Darren Powers  Consumer  United States  Houston  ...  Central
1  Phillina Ober  Home Office  United States  Naperville  ...  Central
2  Phillina Ober  Home Office  United States  Naperville  ...  Central
3  Phillina Ober  Home Office  United States  Naperville  ...  Central
4    Mick Brown  Consumer  United States  Philadelphia  ...  East

   Product ID  Category  Sub-Category \
0  OFF-PA-10000174  Office Supplies  Paper
1  OFF-BI-10004094  Office Supplies  Binders
2  OFF-LA-10003223  Office Supplies  Labels
3  OFF-ST-10002743  Office Supplies  Storage
4  OFF-AR-10003478  Office Supplies  Art

   Product Name  Sales Quantity \
0  Message Book, Wirebound, Four 5 1/2" X 4" Form...  16.448  2
1  GBC Standard Plastic Binding Systems Combs  3.540  2
2  Avery 508  11.784  3
3  SAFCO Boltless Steel Shelving  272.736  3
4  Avery Hi-Liter EverBold Pen Style Fluorescent ...  19.536  3

   Discount  Profit  Profit_Status
0  1.6448  5.5512  Profit
1  0.3540 -5.4870  Loss
2  1.1784  4.2717  Profit
3  27.2736 -64.7748  Loss
4  1.9536  4.8840  Profit

[5 rows x 22 columns]

```

```

In [24]:

# 8. Write a function to return all unique cities.

pd.DataFrame(df['City'].unique(), columns=['City'])

```

Out [24]:

```
      City
0  Houston
1  Naperville
2  Philadelphia
3  Henderson
4  Athens
... ..
537 Whittier
538 Abilene
539 Palatine
540 Port Orange
541 Danbury
542 rows x 1 columns
```

```
      City
0  Houston
1  Naperville
2  Philadelphia
3  Henderson
4  Athens
.. ..
537 Whittier
538 Abilene
539 Palatine
540 Port Orange
541 Danbury

[542 rows x 1 columns]
```

In [28]:

```
# 9. Write a function that returns total Sales and total Profit.

def total_sales_profit(df):
    return df['Sales'].sum(), df['Profit'].sum()

sales, profit = total_sales_profit(df)

print("Total Sales:", sales)
print("Total Profit:", profit)
```

```
stdout:
Total Sales: 2326534.3543
Total Profit: 292296.81460000004
```

In [31]:

```
# 10. Write a function to return average sales per category only for 'West' region.

def avg_sales_west(df):
    return df[df['Region'] == 'West'].groupby('Category')['Sales'].mean()

print(avg_sales_west(df))
```

stdout:

Category

Furniture 354.666299

Office Supplies 118.454679

Technology 416.420079

Name: Sales, dtype: float64